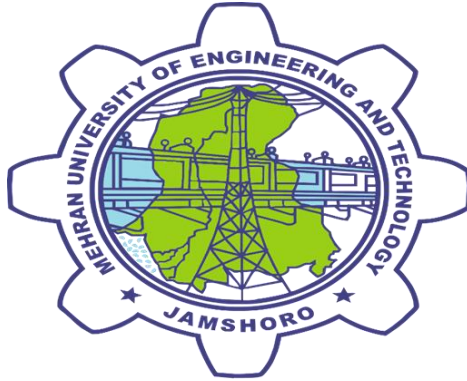




MEHRAN UNIVERSITY OF ENGINEERING AND TECHNOLOGY
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MIT CEP

Comparative Analysis of Intel 8085 Microprocessor and
Intel Core i7-6800K

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MIT 5th Semester 21CS Assignment

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Introduction

Microprocessors play a pivotal role in the evolution of computing technology, shaping the landscape of digital devices and systems. In this essay, we will provide a comparative overview of two prominent microprocessors: the Intel 8085 and the Intel Core i7-6800K. While these processors belong to different eras of computing history, understanding their architectures, features, and capabilities offers valuable insights into the progression of microprocessor technology.

Intel 8085, introduced in 1976, marked a significant milestone in the development of microprocessors. As an 8-bit processor, it represented a foundational building block for early personal computers, embedded systems, and industrial applications. The architecture of the 8085 was relatively simple, featuring a single accumulator and six general-purpose registers. With a clock speed ranging from 2 to 3 MHz, the 8085 executed instructions at a modest pace compared to contemporary processors. Fabricated using NMOS technology, the 8085 typically came in a 40-pin dual in-line package (DIP), reflecting the technological constraints of its time. (1)

The Intel Core i7-6800K, released in 2016 as part of the Broadwell-E lineup, exemplifies the advancements made in microprocessor design and performance over the decades. As a member of the Core i7 series, the i7-6800K targets enthusiasts and power users seeking high-performance computing solutions. Built on a modern 14-nanometer architecture, the i7-6800K features six physical cores with support for Hyper-Threading, allowing it to handle up to 12 threads simultaneously. With a base clock speed of 3.4 GHz and turbo boost capability up to 3.8 GHz, the i7-6800K delivers exceptional processing power for demanding tasks. Furthermore, the processor boasts a sizable L3 cache of 15MB, enhancing its performance in multi-threaded applications. Compatible with DDR4 memory and the LGA 2011-v3 socket, the i7-6800K offers flexibility and scalability for a wide range of computing environments. (2)



Figure 2 CPU Intel 8085



Figure 1 CPU Intel i7-6800k

Timeline of Intel Core i7

The tabular report shares timeline of the Intel Core i7 series, highlighting key releases and advancements from the first generation to the latest generation of the Intel Core i7: [\(3\)](#)

Generation	Release Year	Notable Releases	Architectural Changes/Improvements
1st	2008	Core i7-9xx Bloomfield	Introduction of Nehalem microarchitecture featuring Turbo Boost, Hyper-Threading, and integrated memory controller.
2nd	2011	Core i7-2xxx Sandy Bridge	Enhanced Sandy Bridge microarchitecture with improved performance and energy efficiency. Introduction of AVX instructions and 32nm process technology.
3rd	2012	Core i7-3xxx Ivy Bridge	Implementation of Ivy Bridge microarchitecture featuring improved graphics performance and support for PCIe 3.0.
4th	2013	Core i7-4xxx Haswell	Introduction of Haswell microarchitecture with enhanced CPU and GPU performance, improved power efficiency, and support for DDR3/DDR4 memory.
5th	2015	Core i7-5xxx Broadwell	Implementation of Broadwell microarchitecture featuring improved efficiency and graphics performance, built on 14nm process technology.
6th	2015	Core i7-6xxx Skylake	Introduction of Skylake microarchitecture with focus on performance, power efficiency, and enhanced graphics capabilities.
7th	2017	Core i7-7xxx Kaby Lake	Incremental improvements over Skylake with higher clock speeds, improved 4K video playback, and enhanced security features.
8th	2018	Core i7-8xxx Coffee Lake	Introduction of Coffee Lake microarchitecture with increased core counts, higher clock speeds, and improved gaming performance.
9th	2018	Core i7-9xxx Coffee Lake Refresh	Further enhancements to Coffee Lake microarchitecture with increased core counts, higher clock speeds, and improved thermal management.
10th	2020	Core i7-10xxx Comet Lake	Introduction of Comet Lake microarchitecture with higher core counts, improved clock speeds, and support for faster memory.

Generation	Release Year	Notable Releases	Architectural Changes/Improvements
11th	2021	Core i7-11xxx Rocket Lake (Desktop)	Transition to Cypress Cove microarchitecture with improved IPC, higher clock speeds, and PCIe 4.0 support.
12th	2021	Core i7-12xxx Alder Lake (Desktop)	Introduction of Alder Lake microarchitecture featuring hybrid architecture with performance and efficiency cores, DDR5/DDR4 support, and PCIe 5.0.

The architectural changes and improvements mentioned include advancements in microarchitecture, process technology, core counts, clock speeds, memory support, and integrated graphics performance across different generations of the Intel Core i7 series.

Architecture

Intel 8085 Architecture:

The architecture of the Intel 8085 microprocessor, focuses on its key components, including registers, arithmetic and logic unit (ALU), control unit, and memory interface.

Registers:

1. **Accumulator (A):** The accumulator is an 8-bit register used for arithmetic and logic operations. It stores the result of calculations performed by the ALU and serves as a temporary storage location for data manipulation.
2. **General-Purpose Registers (B, C, D, E, H, L):** The 8085 includes six additional 8-bit general-purpose registers, which can be used for various data manipulation tasks. These registers provide flexibility for programmers to store and manipulate data during program execution.
3. **Program Counter (PC):** The program counter is a 16-bit register that stores the memory address of the next instruction to be executed. It automatically increments after each instruction fetch operation, facilitating sequential program execution.
4. **Stack Pointer (SP):** The stack pointer is a 16-bit register used to manage the stack in memory. It stores the address of the current top of the stack, allowing for efficient storage and retrieval of subroutine return addresses and local variables.

Arithmetic and Logic Unit (ALU):

The ALU is a crucial component of the Intel 8085 microprocessor responsible for performing arithmetic and logic operations on data. The ALU in the 8085 supports a variety of operations, including addition, subtraction, bitwise AND, bitwise OR, bitwise XOR, complementation, and shift operations. It operates on 8-bit binary data and produces results based on the instructions fetched from memory. The ALU communicates with other components of the microprocessor, such as registers and the control unit, to execute instructions efficiently.

Control Unit:

The control unit in the Intel 8085 microprocessor is responsible for coordinating the execution of instructions and controlling the flow of data within the CPU. It decodes instructions fetched from memory and generates control signals to coordinate the activities of other components, such as the ALU, registers, and memory interface. The control unit ensures that instructions are executed in the correct sequence and that data is transferred between registers and memory as needed. It plays a critical role in ensuring the proper functioning of the microprocessor and executing programs accurately.

Memory Interface:

The Intel 8085 microprocessor features a memory interface that enables communication with external memory devices, such as RAM and ROM. The memory interface includes address bus lines, data bus lines, and control signals that facilitate data transfer between the microprocessor and memory. The address bus carries memory addresses generated by the CPU, allowing it to access specific locations in memory. The data bus is used to transfer data between the CPU and memory, enabling read and write operations. Control signals govern the timing and direction of data transfer, ensuring proper synchronization between the microprocessor and memory devices. (4)

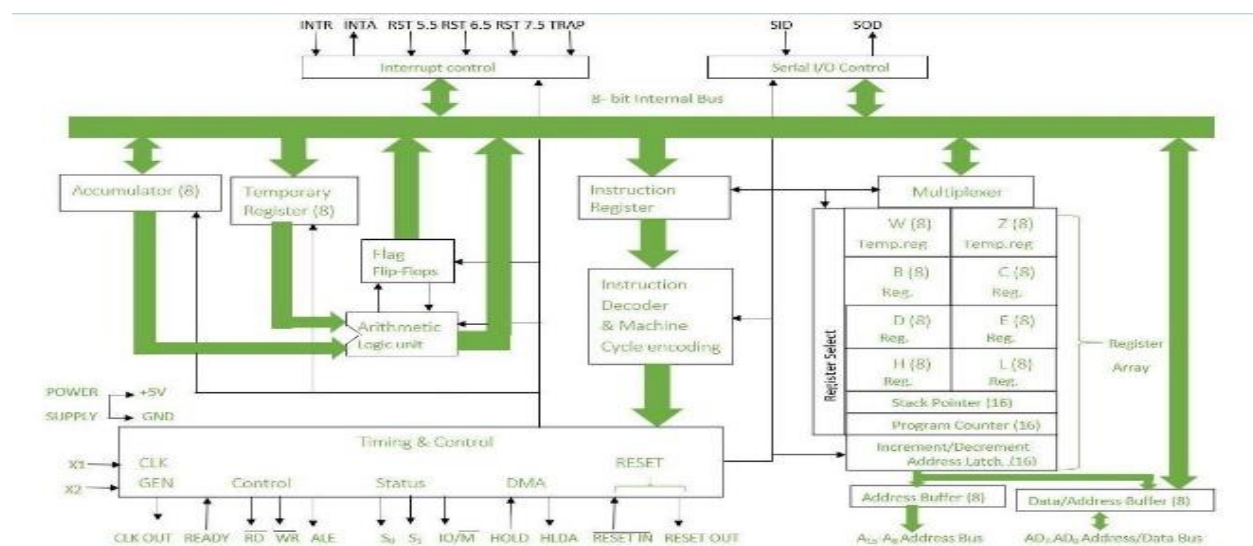


Figure 3 Intel 8085 Architecture

Intel Core i7-6800k Architecture:

Multi-Core Design:

At the heart of the Intel Core i7-6800K is a multi-core design, featuring six physical cores capable of executing simultaneous threads. With support for Hyper-Threading technology, the i7-6800K can handle up to 12 threads concurrently, enabling efficient multitasking and parallel processing. This multi-core architecture allows the processor to divide complex tasks into smaller units and execute them in parallel, significantly improving overall system performance and responsiveness. (5)

Cache Hierarchy:

The cache hierarchy of the Intel Core i7-6800K plays a crucial role in optimizing memory access and reducing latency. The processor features three levels of cache: L1 cache, L2 cache, and L3 cache. The L1 cache is divided into separate instruction and data caches, providing fast access to frequently used instructions and data. The L2 cache serves as a larger buffer between the CPU cores and main memory, storing recently accessed data for quicker retrieval. The L3 cache, shared among all cores, acts as a centralized pool of memory for storing frequently accessed data and instructions, further reducing memory latency and improving performance.

Instruction Set:

The Intel Core i7-6800K supports a wide range of instructions designed to facilitate efficient execution of various computing tasks. It features an extensive instruction set architecture (ISA) that includes support for SIMD (Single Instruction, Multiple Data) instructions, allowing for parallel processing of data across multiple cores. Additionally, the i7-6800K incorporates advanced instructions such as SSE (Streaming SIMD Extensions), AVX (Advanced Vector Extensions), and AES-NI (Advanced Encryption Standard New Instructions), enabling accelerated processing of multimedia, scientific, and encryption algorithms.

Other Relevant Features: In addition to its multi-core design, cache hierarchy, and instruction set, the Intel Core i7-6800K boasts several other relevant features that contribute to its performance and versatility. These include:

- **Turbo Boost Technology:** Dynamic overclocking capability that automatically increases clock speeds to handle peak workloads.
- **DDR4 Memory Support:** Compatibility with DDR4 memory technology, offering increased memory bandwidth and improved efficiency.
- **Enhanced Graphics Performance:** Integration of Intel HD Graphics 530 for improved graphics rendering and multimedia capabilities.
- **PCIe 3.0 Support:** High-speed connectivity with PCIe 3.0 interface for fast data transfer and expansion options. (6)

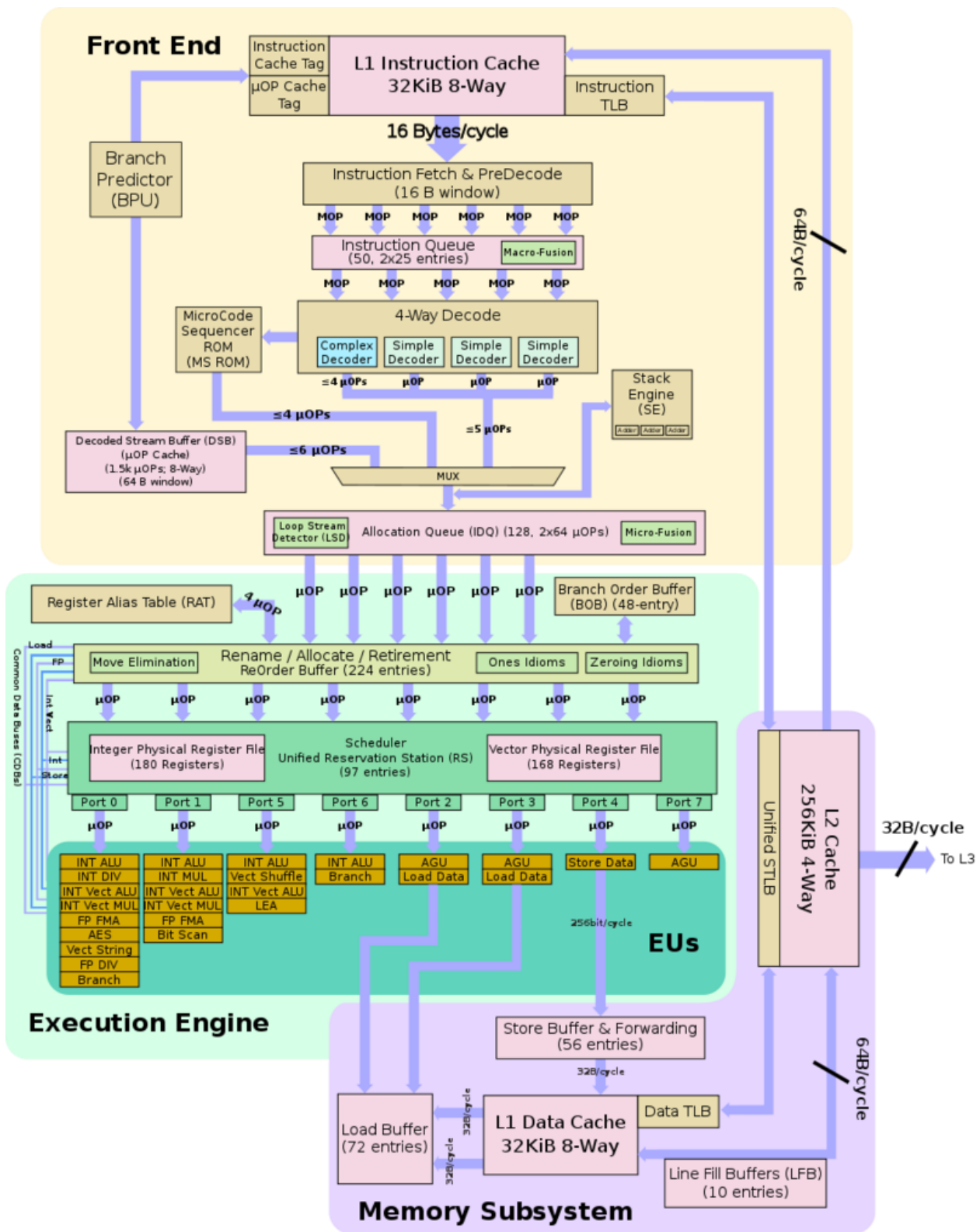


Figure 4 Intel i7-6th gen Broadwell-E Architecture

Architecture Comparison of Intel 8085 Microprocessor and Intel Core i7-6800K:

Differences:

Aspect	Intel 8085 Microprocessor	Intel Core i7-6800K
Microarchitecture	Simple 8-bit microarchitecture optimized for basic tasks	Complex and advanced microarchitecture for high performance computing
Instruction Set	Relatively limited instruction set for basic operations	Extensive instruction set including SIMD, AVX, AES-NI for multimedia, encryption, and scientific computing
Number of Cores	Single-core processor	Six physical cores with hyper-threading (up to 12 threads)
Cache Hierarchy	Limited cache hierarchy with small on-chip cache	Advanced cache hierarchy with multiple levels (L1, L2, L3) for improved performance

Similarities:

Aspect	Intel 8085 Microprocessor	Intel Core i7-6800K
Registers	Essential registers present	Registers such as accumulator, general-purpose, program counter, and stack pointer present
Arithmetic and Logic Unit (ALU)	ALU optimized for 8-bit operations	More complex ALU capable of handling wider data paths and sophisticated operations
Control Unit	Responsible for decoding instructions and coordinating execution	Ensures proper sequencing and data flow within the CPU
Memory Interface	Facilitates communication with external memory devices	Allows data transfer between CPU and memory for read and write operations

Comparing Integration of Microprocessors: Intel 8085 vs Intel Core i7-6800K

Integration Level of the Intel 8085 Microprocessor:

The Intel 8085, introduced in 1976, embodies an early stage of microprocessor technology. Its level of integration reflects the constraints of the era's semiconductor fabrication processes, limiting the extent of components that could be placed on a single chip. Despite this, the 8085 includes fundamental elements such as the ALU, registers, control unit, and memory interface, though it lacks modern features like on-chip cache and integrated graphics. The need for external components to complement its functionality adds complexity to system design.

Integration Level of the Intel Core i7-6800K:

Released in 2016, the Intel Core i7-6800K represents a pinnacle of integration in microprocessor design. Advances in semiconductor technology, notably the 14nm FinFET process, enabled a high level of integration. The i7-6800K encompasses a wide array of features including multi-core processing, sophisticated cache hierarchy, integrated memory controller, PCIe interface, and integrated GPU. Its integration eliminates the need for many external components, streamlining system architecture and reducing power consumption and cost. (7)

Comparison:

1. **Component Integration:** The i7-6800K integrates a broad range of features compared to the 8085, which lacks many modern functionalities.
2. **System Complexity:** The 8085 requires external components, increasing system complexity, whereas the i7-6800K's high integration level reduces complexity.
3. **Performance and Efficiency:** The i7-6800K's advanced features and architecture lead to improved performance, efficiency, and scalability compared to the 8085.

Fabrication Process, Transistor Count, Die Size, and Packaging Technology:

Aspect	Intel 8085	Intel Core i7-6800K
Fabrication Process	3-micron NMOS technology	14nm FinFET technology
Transistor Count	Approximately 6,000 transistors	Approximately 3.2 billion transistors
Die Size	Approximately 34 mm ²	Approximately 242 mm ²
Packaging Technology	40-pin DIP	LGA package

Performance Analysis: Intel 8085 vs Intel Core i7-6800K

Aspect	Intel 8085	Intel Core i7-6800K
Computational Power	The Intel 8085, with its humble 8-bit architecture, serves admirably for simpler computing tasks, akin to the workhorse of its time.	The Intel Core i7-6800K, a modern marvel, boasts a multi-core design and advanced instruction set, catering to today's demanding computing needs.
Speed	The 8085 hums along at clock speeds ranging from 2 to 3.1 MHz, a testament to its era's technological prowess.	In contrast, the i7-6800K blazes through tasks at a base clock speed of 3.4 GHz, with Turbo Boost technology propelling it even further when needed.
Multitasking Capabilities	The Intel 8085's single-core nature limits its multitasking finesse, capable but modest in its approach.	Enter the i7-6800K with its multi-core architecture, seamlessly juggling multiple tasks with finesse, making multitasking a breeze.

Comparison of Energy Consumption Metrics: Intel 8085 vs Intel Core i7-6800K

Aspect	Intel 8085	Intel Core i7-6800K
Thermal Design Power (TDP)	The 8085's modest TDP, typically ranging from 1 to 3 watts, speaks to its efficient yet capable nature.	The i7-6800K, with its higher TDP of 140 watts, showcases the power needed to fuel its exceptional performance.
Power Efficiency	Though the 8085 shines in its era for balanced performance and power consumption, it pales in comparison to the i7-6800K's optimized power management, a testament to modern engineering marvels.	The i7-6800K's power management prowess optimizes energy consumption, striking a harmonious balance between performance and efficiency.

Conclusion

The comparative analysis underscores stark differences between the Intel 8085 microprocessor and the Intel Core i7-6800K, revealing significant advancements in microprocessor technology over the years.

Key Findings: The Intel Core i7-6800K showcases superior performance, speed, and multitasking capabilities compared to the Intel 8085. Its multi-core architecture, higher clock speeds, and advanced instruction set contribute to its dominance in modern computing tasks.

Strengths and Weaknesses: The Intel 8085 boasts simplicity and low power consumption, but its limited performance and lack of multitasking capabilities make it obsolete for modern computing. In contrast, the i7-6800K excels in performance and versatility but consumes more power and is more complex.

Recommendations and Predictions: For contemporary computing tasks, the Intel Core i7-6800K is recommended due to its high performance and multitasking abilities. Future microprocessor developments are anticipated to focus on enhancing performance, power efficiency, and integration, with advancements such as smaller transistor sizes and specialized processing units becoming more prevalent.

References

- (1) – (Book) Barry B. Brey – The Intel Microprocessors
https://books.google.com.pk/books?hl=en&lr=&id=gr0OZ3YPTYwC&oi=fnd&pg=PA1&dq=Brief+overview+of+Intel+8085+microprocessor&ots=HDOqVBuonu&sig=15TcRqrxFcNaEcTzs582uVHZOI&redir_esc=y#v=onepage&q=Brief%20overview%20of%20Intel%208085%20microprocessor&f=false
- (2) – (Article) Jarred Walton – Intel Core i7-6800K Review
<https://www.pcgamer.com/intel-core-i7-6800k-review/>
- (3) – (Webpage) Wikipedia – List of Intel Core i7 Processors
https://en.wikipedia.org/wiki/List_of_Intel_Core_i7_processors
- (4) – (Article) Brian Mwambia - Exploring the Architecture of the 8085 Microprocessor
<https://medium.com/@brianmwambia3/exploring-the-architecture-of-the-8085-microprocessor-a-classic-tale-of-computing-evolution-50373cf6951a>
- (5) – (YouTube Video) Iceberg Tech – The i7-6800k was ahead of its time
https://www.youtube.com/watch?v=zqo_45AUXmA&pp=ygUWaTctNjgwMGsgYXJjaGl0ZWN0dXJlIA%3D%3D
- (6) – (Webpage) Wikipedia – Architecture changes compared to Broadwell microarchitecture
[https://en.wikipedia.org/wiki/Skylake_\(microarchitecture\)#Architecture_changes_compared_to_Broadwell_microarchitecture](https://en.wikipedia.org/wiki/Skylake_(microarchitecture)#Architecture_changes_compared_to_Broadwell_microarchitecture)
- (7) – (Webpage) Tech Power Up - Intel Core i7-6800K
<https://www.techpowerup.com/cpu-specs/core-i7-6800k.c1844>

Department of Computer Systems Engineering
Mehran University of Engineering and Technology, Jamshoro

Course: Microprocessor and Interfacing (CS-311)

Instructor	Engr. Athar Ali	Assignment Type	Complex Engineering Problem
Semester	5 th	Year	3 rd
Submission Deadline	04 – 04 - 2024	Assessment Score	_____10_____

Complex Computing Problem - Characteristics

1	Range of Conflicting Requirements	☒
2	Depth of Analysis Required	☒
3	Depth of knowledge Required	☒
4	Familiarity of Issues	☒
5	Level of Problem	☐
6	Extent of Stakeholder involvement and level of Conflicting Requirements	☐
7	Consequences	☒
8	Independence	☐

Problem Description

Explore the architecture, integration and techniques of modern CPUs and compare their specification with 8085 or 8086 processors of intel. Keep following parameters while attempting the assignment.

1. How intensive integration of modern CPUs impacts system performance and energy consumption of computer system.
2. Write a technical report of your findings in the context of research.
3. Refer and cite scholarly research articles based on trade-off between modern CPUs and classical CPUs.
4. Cite at least 3 sources in APA referencing format

Rubrics	Assessment					Marks
	Unaccepta ble	Poor	Acceptable	Adequate	Proficient	
R1 Identification of constraints/requirements/demands/ research gap or challenges well defined	☐	☐	☐	☐	☐	1
R2 knowledge (standards)	☐	☐	☐	☐	☐	1
R3 Familiarity of issues	☐	☐	☐	☐	☐	2
R4 Critical Analysis and problem understanding	☐	☐	☐	☐	☐	2
R5 Efficiency of the solution	☐	☐	☐	☐	☐	2
R6 Technical Writing	☐	☐	☐	☐	☐	2
Total Marks						10